r/Futurology

Is the world heading toward a massive electricity shortage? EVs and AI supercomputers seem unsustainable

Posted by u/Tall-Caterpillar-641 • • 0 points • 72 comments

So I've been thinking about this a lot lately and wanted to get everyone's perspective.

From what I understand, we're already in a situation globally where electricity generation is tight demand is barely being met in a lot of regions. But now we're rapidly shifting to:

1. Electric vehicles-Governments are pushing EVs hard, which means millions of cars that previously ran on gas will now need to plug into the grid.
2. AI datacenters-I have heard that the supercomputers needed for AI training and operations require absolutely massive amounts of electricity potentially more than entire countries current usage

This seems like we're heading toward a perfect storm. so i want to know

* Are we actually short on electricity globally right now or is generation keeping up with demand?
* What solutions are countries seriously pursuing? Is nuclear the only realistic option to scale fast enough? Solar and wind are great but can they really meet this exponential demand growth?
* What happens when AI systems become more widespread? If AI really does take over more industries and processes, won't the electricity demands become completely unsustainable?

I'm genuinely curious if there's a realistic path forward here or if we're just ignoring a looming infrastructure crisis. Are there breakthrough technologies in generation or storage that could actually solve this?

Would love to hear from anyone who works in energy, policy, or has done deep research on this.

Comments

u/avatarname • 1 points • 2025-11-12 09:13 UTC

"Is nuclear the only realistic option to scale fast enough? Solar and wind are great but can they really meet this exponential demand growth?"

It is actually vice versa, solar and wind can scale much faster than nuclear if you have a lot of land at least and no complicated bureaucratic procedures. Of course these are two sometimes big IFs but nuclear still takes a lot of time to build up, even if you could cut bureaucracy, especially in countries that did not have nuclear before. Case in point in Estonia they want to build a small modular reactor and they have been talking about it for few years already and even if they get funding and all, it will be up maybe by 2033. In that time you can build out a whole bunch of solar and wind. Of course if we talk about a giant nuclear plant the maths is different but it then could take even more time or for a small country like Estonia it is not actually financially wise who would finance it as at the moment energy demand is not there and by the time it is done, there will be way more solar, wind and cheaper batteries.

u/DueAnnual3967 • 1 points • 2025-11-11 22:40 UTC

I think there is too much crying about lack of this and that, like some years ago, less than 10 years ago they said Bitcoin would drain massive amounts of power. It has not happened. I calculated that to run all passenger vehicles we would need to add approx 26% more electricity to grid. Which would seem bad if we weren't in track to add over 40% by 2029. Some of that will replace natural gas.

u/Scope_Dog • 2 points • 2025-11-08 18:56 UTC

No the world is not headed toward an electricity shortage. We are building massive, massive amounts of renewable power. And the pace is only increasing. It took over a hundred years for the world to build out its fossil fuel power infrastructure to what it is. It will take a fraction of that time for it all (plus way more)to be replaced by renewable sources. Keep in mind that renewables are getting cheaper and better every year and deployment speeds are increasing exponentially as well. Technological change is not linear. Look no further than that little device in your pocket for proof.

u/bobjoylove • 0 points • 2025-11-08 18:08 UTC

People ignore the small shit. Like most of the transformers are now sourced from China. Trade wars could easily see this essential building block become a major issue in building out the grid.

u/Duckbilling2 • 1 points • 2025-11-08 17:31 UTC

In the USA, AI data centers will be a problem with the grid unless something changes. Also with

EVs are not an issue, but we shall see - in the transition to commercial semi trucks it may present a problem

u/johnp299 • 1 points • 2025-11-08 17:21 UTC

Last time I checked, gas stations also use electricity. Pumping, lighting, etc. Same goes for the infrastructure to get gas to the station. And the refineries and so on.

u/Antipolemic • 1 points • 2025-11-08 15:49 UTC

The risk is not one of capacity, but of timing. Projected US demand for data centers/AI is 426 TWh by 2030 (thanks, ironically, to an AI search), which is 133% higher than 2024 levels. Total 2023 US electricity production was 4,178 TWh with breakout as follows: fossil fuels 60% (2/3 of this natural gas), Nuclear (18.5%), renewables (21.4%). With an "all-of-the-above" strategy (excepting coal, perhaps), with heavy focus on renewables projects which can be brought on line relatively quickly (2 year lead times), we can handle the demand. We do have a problem in the US with dilapidated and inadequate transmission networks and sufficient battery storage has not yet fully been realized. However, this problem can be worked around by decentralizing generation and locating renewables closer to their loads. Many tech companies have even created privately owned renewables projects to feed specific data centers. As far as cleaner energy goes, small modular nuclear fission tech offers an attractive and scalable base load option, although significant use of these is still at least 10-20 years off. The only US-based project planned will come on line in in 2032ish time frame I believe. There are only two currently producing SMRs in the world - one in China and one in Russia. In the interim, then, it's going to be renewables and natural gas fired plants as the leading source of generation growth. Traditional nukes will not be built in the US again due to cost, lack of expertise and skilled labor (which leads to massive delays and cost overruns) and it is highly unlikely coal will either (for very good reasons). It's by no means an unsolvable problem.